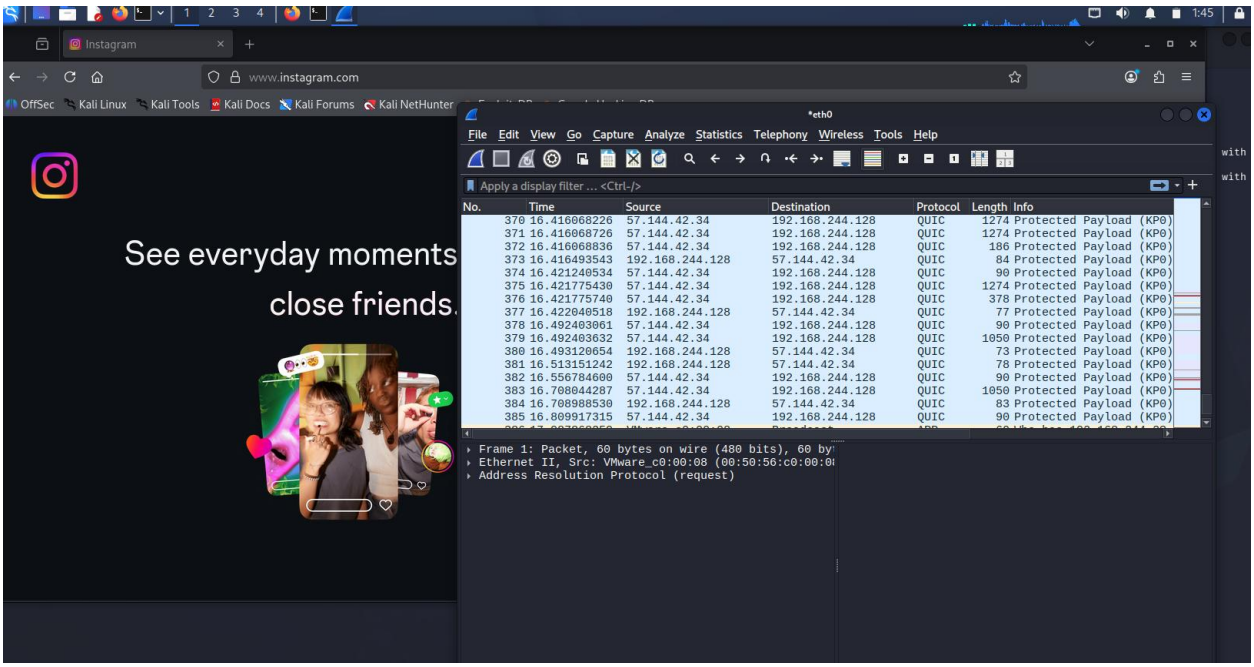
1.Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.



I opened **Wireshark** and selected my main network interface, **eth0**. Then I started capturing packets and opened the Instagram website in **Firefox** by visiting www.instagram.com**.**

While Instagram was loading, **Wireshark captured the network** traffic between **my laptop** and **Instagram's servers.** After the page loaded, I stopped the capture and checked the packet list.

The main protocols visible in the capture were:

QUIC — encrypted web traffic used by Instagram/Google Chrome-style web connections over UDP.

ARP (Address Resolution Protocol) — used on the local network to find the hardware address of devices, such as the router.

QUIC packets exchanged between the local IP address 192.168.244.128 and the remote IP address 57.144.42.34, which is part of the Instagram-related traffic.

2.Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

From the Wireshark capture, I selected three packet/ protocol examples and mapped them to the OSI model:

Packet / Protocol	OSI Layer	Reason
QUIC packet	**Transport layer (Layer 4)**	QUIC transports the encrypted Instagram web data over UDP. It handles connection and data delivery functions similar to TCP, so it is mainly associated with the Transport layer.
UDP *(inside QUIC traffic)*	**Transport layer (Layer 4)**	QUIC uses UDP as its underlying transport protocol to send packets between the laptop and Instagram's server.
ARP request	**Data Link layer (Layer 2)**	ARP is used in the local network to find the MAC address connected to an IP address, such as the router's MAC address. It works with Ethernet frames.

3.Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



4.Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint: Think about how your message is typed, packaged, sent over the internet, and received.

Example: I type "Hello" and send it on WhatsApp.

1. **Application Layer (Layer 7)**
   type "Hello" in WhatsApp and press send.
2. **Presentation Layer (Layer 6)**
   The message is formatted and **encrypted** for secure communication.
3. **Session Layer (Layer 5)**
   It manages the **communication session** between my phone and the receiver's phone.
4. **Transport Layer (Layer 4)**
   The message is broken into **smaller parts** and prepared for delivery.
5. **Network Layer (Layer 3)**
   IP addresses are used to find the destination.
6. **Data Link Layer (Layer 2)**
   The data is put into frames and sent through Wi-Fi or Ethernet.
7. **Physical Layer (Layer 1)**
   The data travels as **radio/electrical signals** through the network.